

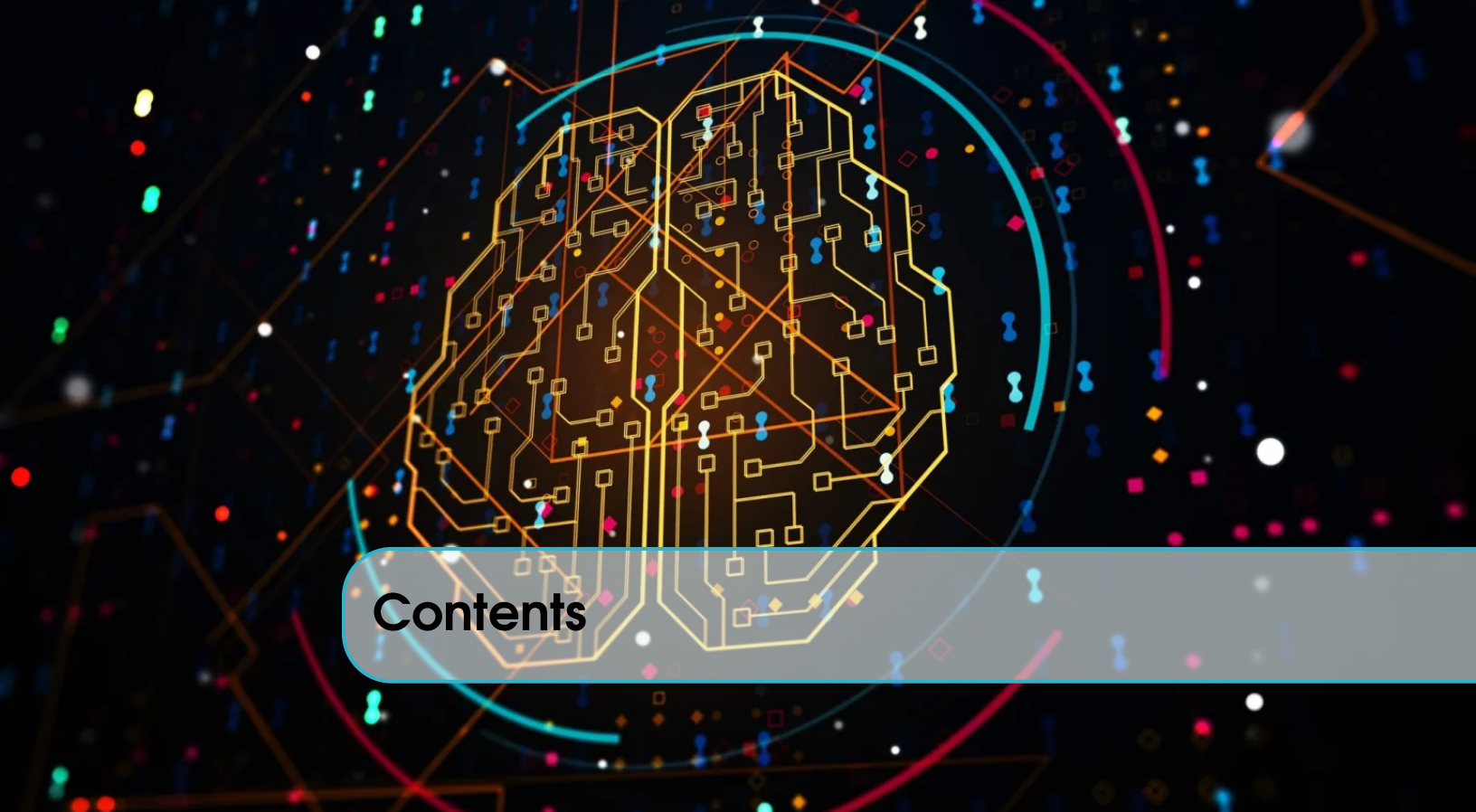
The background of the entire cover is a vibrant, multi-colored cosmic nebula. It features swirling clouds of gas and dust in shades of deep blue, magenta, orange, and yellow. Several bright stars are visible, some with prominent diffraction spikes. The overall effect is one of deep space and celestial wonder.

Advanced Machine Learning

La Voglia di Morire

Vincenzo Salvatore

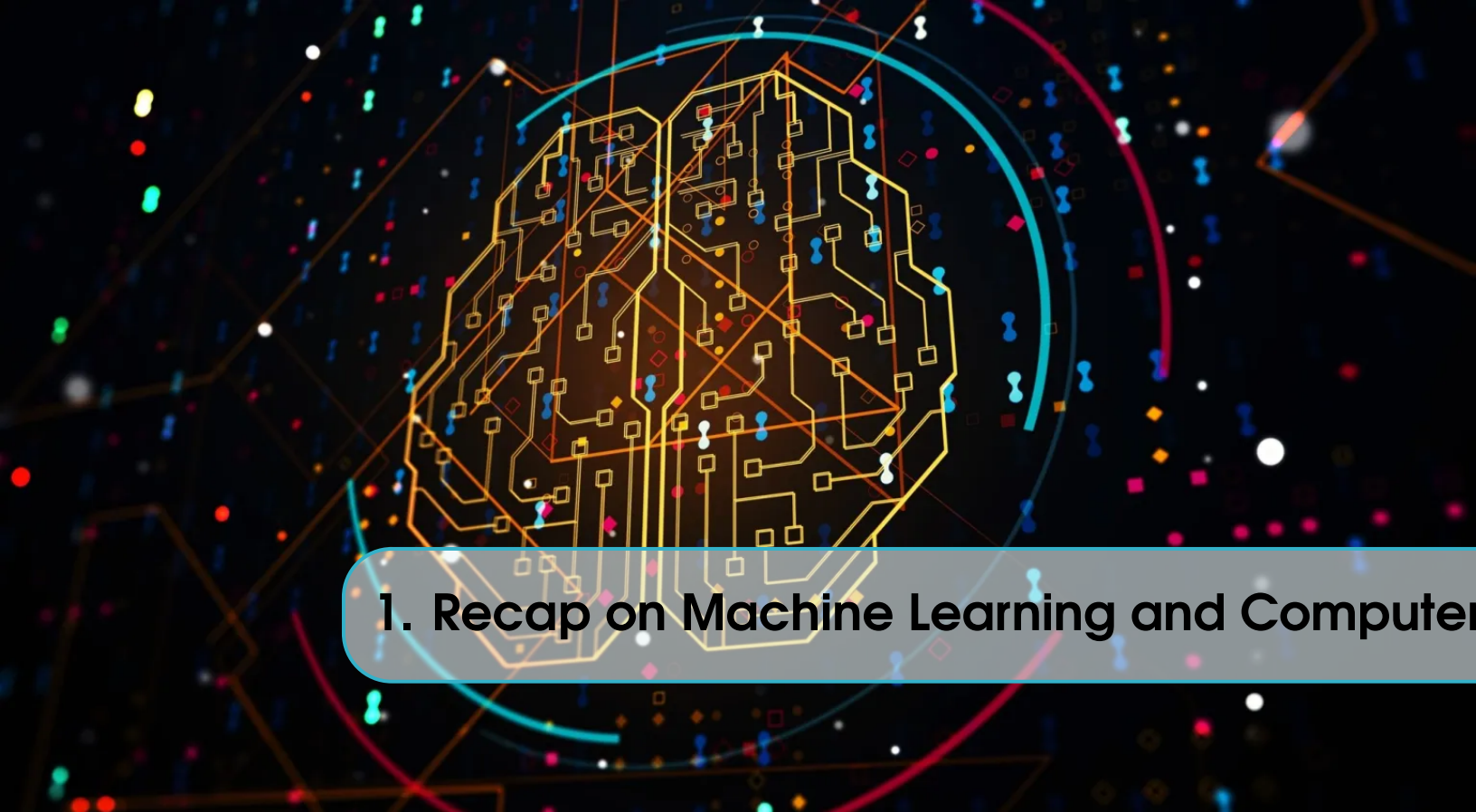
Raffaele Mercurio



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1. Recap on Machine Learning and Computer

1.1 Introduction to the Core Concepts

The contents of this chapter are largely based on the introductory slides of the course [1].

Key Word

- According to Arthur Samuel (1959), **Machine Learning** is the field of study that gives computers the ability to learn without being explicitly programmed.
- Tom Mitchell (1998) defined the well-posed learning problem: a computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .

Machine learning is heavily applied in **Computer Vision**, which can be viewed from three main perspectives: as a science (exploring the computational model of human perception), as engineering (building systems that perceive the world), and through its applications.

Example

Some typical **Computer Vision Applications** include:

- **Medical imaging:** Supporting medical diagnosis and visualization.
- **Surveillance:** Tracking people in airports or train stations.

- **Autonomous driving:** Lane-keeping and pre-crash intervention.

1.2 Image Classification

A core task in computer vision is **Image Classification**: assuming a given set of discrete labels (e.g., {dog, cat, truck, plane}), the goal is to map an input digital image to one of these labels.

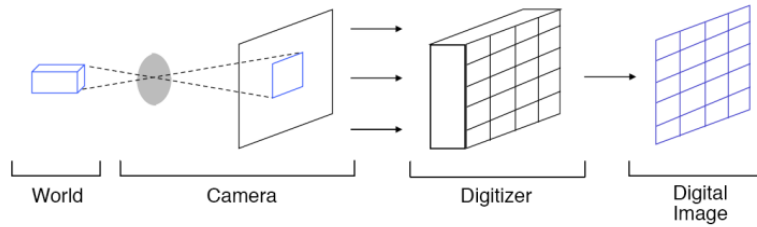


Figure 1.1: Illustration of the (pinhole) camera imaging process.

Note

The Imaging Process: An image is typically formed via a (pinhole) camera model and then passed through a digitizer to obtain a digital image, which the computer interprets as a large multi-dimensional array of numbers.

1.3 Parametric Models and Gradient-Based Optimization

The majority of machine learning models can be seen as **Parametric Models**. A parametric model takes an input x and uses a set of parameters (or weights) W and biases b to produce an output prediction z :

$$z = f(x, W) = Wx + b$$

While the exact nature of the prediction depends on the task, what unifies these models is the **Optimization Process**. We define a Loss function L that measures the error of our predictions, and we iteratively adjust the weights using **Gradient Descent**:

$$W^{t+1} = W^t - \alpha \nabla_W L$$

where α is the learning rate.

By simply changing the *activation function* applied to the linear score z and the corresponding *loss function*, we obtain different classical models: Linear Regression, Logistic Regression, and the Softmax Classifier.

1.3.1 Linear Regression ($f : \mathbb{R}^D \rightarrow \mathbb{R}$)

Linear Regression is used when the target variable is continuous. We use the identity activation function (i.e., no activation), so the predicted output $\hat{y}$ is exactly the linear score:

$$\hat{y} = z = Wx + b$$

We optimize this model by minimizing the **Mean Squared Error (MSE)** loss:

$$L = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

1.3.2 Logistic Regression ($f : \mathbb{R}^D \rightarrow [0, 1]$)

Logistic Regression is used for binary classification. We map the linear score z to a probability between 0 and 1 using the **Sigmoid** activation function $\sigma(z)$:

$$P(Y = 1|X = x) = \sigma(z) = \frac{1}{1 + e^{-z}}$$

The appropriate loss function for this model is the **Binary Cross-Entropy Loss (BCE)**. For a dataset of N samples, it is defined as:

$$L = -\frac{1}{N} \sum_{i=1}^N \left[\underbrace{y_i \log(\sigma(z_i))}_{\text{Active if } y=1} + \underbrace{(1 - y_i) \log(1 - \sigma(z_i))}_{\text{Active if } y=0} \right]$$

Note

Understanding the BCE Formula: This equation uses a clever mathematical trick to combine two mutually exclusive cases into a single line:

- When the true class is $y_i = 1$, the second term $(1 - 1) = 0$ disappears, and the loss evaluates only $-\log(\sigma(z_i))$, penalizing the model if it predicts a low probability.
- When the true class is $y_i = 0$, the first term 0 disappears, and the loss evaluates only $-\log(1 - \sigma(z_i))$, penalizing the model if it predicts a high probability for class 1.

1.3.3 Softmax Classifier ($f : \mathbb{R}^D \rightarrow \{0, \dots, K\}$)

The Softmax Classifier generalizes Logistic Regression to multiple, mutually exclusive classes. It maps the raw scores z into a valid probability distribution over the classes using the **Softmax function**:

$$P(Y = k|X = x_i) = \frac{e^{z_k}}{\sum_j e^{z_j}}$$

Note

Why Softmax? We want to interpret raw classifier scores as *probabilities*. The Softmax function transforms scores such that they are strictly positive and sum to 1.

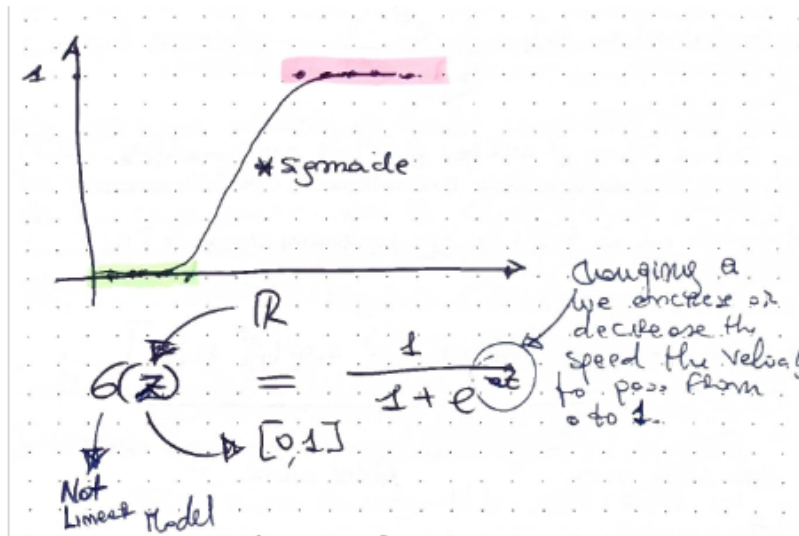


Figure 1.2: Illustration of the Sigmoid function in Logistic Regression.

Example

For an input image of size $32 \times 32 \times 3$ (e.g., CIFAR-10), the flattened array x has 3072 numbers. If we are classifying into 10 categories, the weight matrix W will have dimensions 10×3072 , producing a 10×1 vector of raw class scores before the Softmax is applied.

We then optimize the model using the **Categorical Cross-Entropy Loss**, often referred to as the Negative Log-Likelihood (NLL). Over the entire dataset of N samples, expanding the Softmax probability formula, the loss becomes:

$$NLL(W, X) = - \sum_{i=1}^N \left[\underbrace{W_{y_i}^T x_i}_{\text{Punteggio della classe giusta}} - \underbrace{\log \left(\sum_{k=1}^K \exp(W_k^T x_i) \right)}_{\text{Somma dei punteggi delle altre classi}} \right]$$

Note

Understanding the Softmax NLL Formula: By looking at the expanded formula, we see that the optimizer tries to maximize the linear score $W_{y_i}^T x_i$ of the *correct class* y_i , while simultaneously pushing down the log-sum of the scores of *all available classes*. This creates a competitive environment where increasing the score of the correct class implicitly decreases the relative probability of the others.

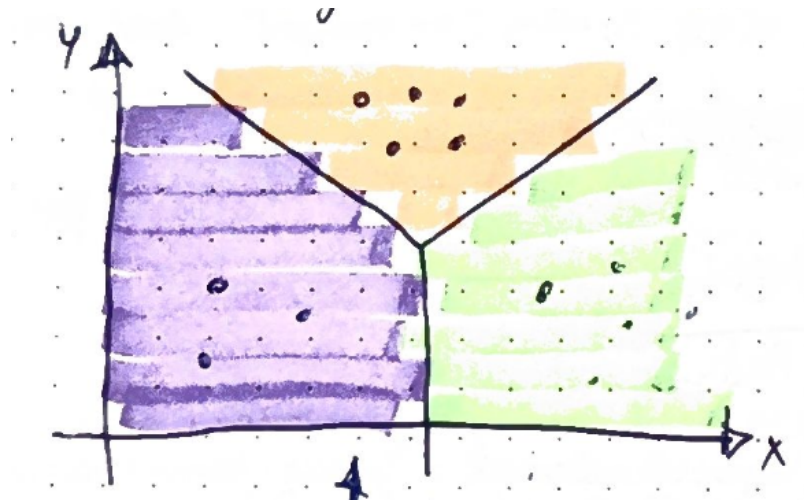


Figure 1.3: Illustration of the Sigmoid function in Logistic Regression.

1.4 Advanced Optimization and Regularization

1.4.1 Regularization

The primary goal of Regularization is to prevent the model from becoming too complex and overfitting the training data (i.e., we use it to combat **High Variance**). We achieve this by adding a penalty term $R(W)$, based only on the model's parameters, directly to the Data Loss.

The new Total Loss becomes:

$$L_{total}(W, X) = \underbrace{L(W, X)}_{\text{Data Loss}} + \underbrace{\lambda R(W)}_{\text{Regularization Term}}$$

Note

When and how is it applied?

Regularization is applied *during training*. By redefining the Total Loss, we also change its gradient. When we apply Gradient Descent, the algorithm must now minimize both the data error AND the complexity of the weights. For example, the gradient of an L2 penalty λW^2 is $2\lambda W$, meaning Gradient Descent will actively shrink the weights at every step (Weight Decay).

The hyperparameter λ acts as the “manipulator” of complexity:

- $\lambda = 0$: We ignore complexity. The model relies entirely on Data Loss (Risk of **Overfitting**).
- $\lambda \rightarrow \infty$: The model becomes “lazy”. The penalty is so high that Gradient Descent brings all weights to zero (Risk of **Underfitting**).

The two most classic mathematical formulations are:

- **L2 Regularization (Ridge)**: $R(W) = \sum W^2$. It punishes larger weights more severely, forcing the network to distribute the “weight” across all features rather than relying heavily on just a few. It shrinks weights but rarely brings them exactly to zero.
- **L1 Regularization (Lasso)**: $R(W) = \sum |W|$. It punishes weights constantly. This has the mathematical effect

of bringing less important weights exactly to zero, acting as an automatic *Feature Selection* mechanism.

Note: In deep neural networks, we also use architectural regularizers like Dropout and Batch Normalization.

1.4.2 Stochastic Gradient Descent (SGD)

From a theoretical standpoint, our objective is to minimize the total loss over the entire dataset, which includes both the data loss and the regularization term:

$$L(W) = \frac{1}{N} \sum_{i=1}^N L_i(x_i, y_i, W) + \lambda R(W)$$

To update the weights, we need the exact gradient of this total loss:

$$\nabla_W L(W) = \frac{1}{N} \sum_{i=1}^N \nabla_W L_i(x_i, y_i, W) + \lambda \nabla_W R(W)$$

Computing this true gradient (the **Full Batch** gradient) requires evaluating the sum over all N samples in the dataset at every single step. When N is in the millions, this becomes computationally unfeasible.

Key Word

Stochastic Gradient Descent (SGD): Instead of the full sum, SGD approximates the true gradient by computing the sum over a small **minibatch** of examples (commonly 32, 64, or 128). This introduces noise but allows for much faster and frequent parameter updates.

1.5 Backpropagation and Neural Networks

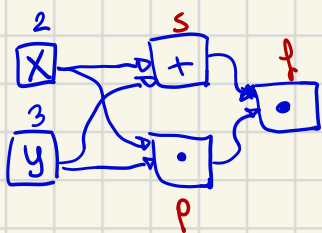
1.5.1 Backpropagation

To update weights, we compute gradients via **Backpropagation**, which applies the chain rule from the output back to the inputs (backward pass).

Note

Backprop with Matrices: When inputs and weights are vectors or matrices, local gradients become *Jacobian matrices*. However, Jacobians are often sparse. Instead of explicitly forming them, we use implicit multiplication. A very useful trick to remember: the downstream gradient dL/dx **always has the same shape** as x !

Here are some solved exercises demonstrating the Forward and Backward pass step-by-step:



FORWARD PASS

$$s = x + y = 2 + 3 = 5$$

$$p = x \cdot y = 2 \cdot 3 = 6$$

$$f = s \cdot p = 30$$

Reverse Pass

$$f = s \cdot p$$

$$p = x \cdot y$$

$$\frac{\partial f}{\partial s} = p$$

$$\frac{\partial f}{\partial p} = s$$

$$\frac{\partial p}{\partial x} = y$$

$$\frac{\partial p}{\partial y} = x$$

$$s = x + y$$

$$\frac{\partial s}{\partial x} = 1 + 0 = 1$$

$$\frac{\partial s}{\partial y} = 0 + 1 = 1$$

Applico la chain rule

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial s} \cdot \frac{\partial s}{\partial x} + \frac{\partial f}{\partial p} \cdot \frac{\partial p}{\partial x} = p \cdot 1 + s \cdot y = 6 \cdot 1 + 5 \cdot 3 = 21$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial s} \cdot \frac{\partial s}{\partial y} + \frac{\partial f}{\partial p} \cdot \frac{\partial p}{\partial y} = p \cdot 1 + s \cdot x = 6 \cdot 1 + 5 \cdot 2 = 16$$

Cosa capisco da questo esercizio? Capisco quanto in peso x e y impattano rispetto alla funzione di Loss f . Vedo che x ha un impatto maggiore.

ESERCIZIO 3

Consider a 2-Layer function: $z = \tanh(xy + 1)$

$$x = 1 \text{ e } y = 2$$

• Build a computation graph:

• Forward pass: compute z .

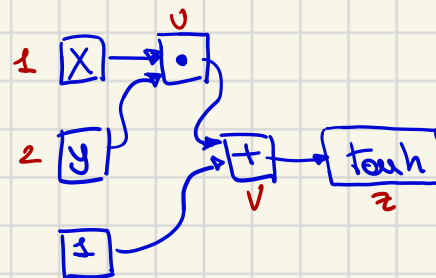
• Reverse pass: compute gradient $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$.

FORWARD:

$$u = x \cdot y = 2$$

$$v = u + 1 = 3$$

$$z = \tanh(v) = \tanh(3) \approx 0.995$$



Reverse Pass

$$z = \tanh(v) \leadsto \frac{\partial z}{\partial v} = \text{sech}^2(v) = 1 - \tanh^2(v) \approx 1 - (0.995)^2 \approx 0.01$$

$$v = u + 1 \leadsto \frac{\partial v}{\partial u} = 1 \quad \bullet \quad u = x \cdot y \leadsto \frac{\partial u}{\partial x} = y = 2$$

• Chain Rule

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial u} \cdot \frac{\partial u}{\partial x} = 0.01 \cdot 1 \cdot 2 \approx 0.02$$

$$\frac{\partial u}{\partial y} = x = 1$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial u} \cdot \frac{\partial u}{\partial y} = 0.01 \cdot 1 \cdot 1 \approx 0.01$$

ESERCIZIO 4 given: $g(w) = (w-3)^2$ $w_0 = 0$ $\alpha = 0.1$

$$g'(w) = 2(w-3)$$

$$w^k = w^{k-1} - \alpha \nabla g(w^{k-1})$$

- ① WRITE the gradient descent Rule
- ② Compute the first 5 iter... $w_1 \dots w_5$
- ③ sketch the cost function values.

$$w_0 = 0$$

$$w_1 = 0,8 \cdot 0 + 0,6 = 0,6$$

$$w_2 = 0,8 \cdot 0,6 + 0,6 = 1,08$$

$$w_3 = 0,8 \cdot 1,08 + 0,6 = 1,464$$

$$w_4 = 0,8 \cdot 1,464 + 0,6 = 1,7712$$

$$w_5 = 0,8 \cdot 1,7712 + 0,6 = 2,01696$$

Ad ogni passo d'iterazione ci spinge un po' più vicino al minimo vero $w^* = 3$. La velocità con cui ci avviciniamo a 3 diminuisce gradualmente.

$$w_{k+1} = w_k - \alpha g'(w_k) =$$

$$w_{k+1} = w_k - 0,2(w_k - 3) =$$

$$w_{k+1} = w_k - 0,2w_k + 0,6$$

$$w_{k+1} = (1 - 0,2)w_k + 0,6$$

$$w_{k+1} = 0,8w_k + 0,6$$

Notiamo come solo lo 0,2% del vecchio valore è mantenuto.

Esercizio 5

$$g(w_1, w_2) = w_1^2 + 2w_2^2$$

① Compute the gradient $\nabla g(w)$

② Apply GD from initial point $(w_1, w_2) = (2, 1)$ with learning rate $\alpha = 0,2$

③ Compute the first 3 update explicitly.

for computing the gradient $\nabla g(w_1, w_2)$ we have to apply the first derivate to the function $g(w_1, w_2) \leadsto g'(w_1, w_2) = 2w_1 + 4w_2$

$$\nabla g(w_1, w_2) = \begin{bmatrix} 2w_1 \\ 4w_2 \end{bmatrix}$$

re gradiente è la derivata della funzione.

Rule for each weight:

$$w_1^{k+1} = w_1^k - \alpha \nabla g(w_1) =$$

$$w_1^{k+1} = w_1^k - 0,2(2w_1^k) =$$

$$w_1^{k+1} = w_1^k - 0,4w_1^k =$$

$$w_1^{k+1} = 0,6w_1^k$$

$$w_2^{k+1} = w_2^k - \alpha \nabla g(w_2) =$$

$$w_2^{k+1} = w_2^k - 0,2(4w_2^k) =$$

$$w_2^{k+1} = w_2^k - 0,8w_2^k =$$

$$w_2^{k+1} = 0,2w_2^k$$

Questo è il modo in cui cominceremo i pesi.

Let's compute the iteration starting with $(2, 1)$

$$k=1 \leadsto (w_1, w_2) = (0,6 \cdot 2, 0,2 \cdot 1) = (1,2, 0,2)$$

$$k=2 \leadsto (1,2, 0,2) = (0,72, 0,04)$$

$$k=3 \leadsto (0,72, 0,04) = (0,432, 0,008)$$

Obiettivo: Vogliamo capire come la funzione di Loss guida e aggiornare dei pesi per far sì che la nostra rete produce output sempre più vicini a quelli desiderati.

EX.

Immagina di avere un modello lineare usiamo ad esempio $(X, y) = (2, 5)$

$$\hat{y} = w \cdot x + b$$

peso
Bias
input
output Predetto.

PASSO 1: Inizializzo i PESI

Pesi iniziali: $w = 0,5 \quad b = 0,0$

! Non so più nulla quindi impostiamo valori casuali.

PASSO 2: FORWARD PASS (PREDIZIONE)

$$\hat{y} = w \cdot x + b = 0.5 \cdot 2 + 0 = 1 \quad \leftarrow \text{PREDIZIONE}$$

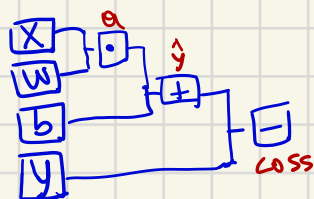
PASSO 3: calcolo delle LOSS (ERRORE). usiamo MSE

$$L = \frac{1}{2}(\hat{y} - y)^2 = \frac{(1 - 5)^2}{2} = \frac{16}{2} = 8 \quad \leftarrow \text{siamo molto lontani dal TARGET.}$$

PASSO 4: Backpropagation (calcolo del Gradiente): come cambiamo i Pesì??

▼ Deriviamo la Loss Rispetto a $\hat{y}$

GRAFO COMPUT:



$$\begin{cases} a = w \cdot x \\ \hat{y} = a + b \\ L = \frac{1}{2}(\hat{y} - y)^2 \end{cases}$$

BACKWARD PASS

$$\bullet \frac{\partial L}{\partial \hat{y}} = \hat{y} - y$$

$$\bullet \frac{\partial \hat{y}}{\partial a} = 1$$

$$\bullet \frac{\partial a}{\partial w} = x$$

$$\bullet \frac{\partial \hat{y}}{\partial b} = 1$$

$$\boxed{w} \rightarrow \frac{\partial L}{\partial w} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial a} \cdot \frac{\partial a}{\partial w} = (\hat{y} - y) \cdot x = (1 - 5) \cdot 2 = \boxed{-8}$$

$$\boxed{b} \rightarrow \frac{\partial L}{\partial b} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial b} = (\hat{y} - y) \cdot 1 = \boxed{-4}$$

PASSO 5: GD update dei Pesì. $\alpha = 0.1$

Regola di aggiornamento: $w := w - \alpha \frac{\partial L}{\partial w}$

$$b := b - \alpha \frac{\partial L}{\partial b}$$

$$\begin{aligned} w^1 &= 0.5 - 0.1 \cdot (-8) = 1.3 \\ b^1 &= 0 - 0.1 \cdot (-4) = 0.4 \end{aligned} \quad \text{NUOVI PESI} \quad k=1$$

PREDICT

$$\hat{y} = w^1 x + b^1 = 1.3 \cdot 2 + 0.4 = 3 \quad \text{Nota come la predizione diventa più precisa e vicino al 5 del Dataset.}$$

Orizzontalmente questo Processo viene eseguito su ogni riga del Dataset. Non sulle stesse.

1.5.2 Deeper Neural Networks

By stacking multiple layers and applying non-linear activation functions, we create multi-layer Neural Networks (e.g., $f = W_2 \max(0, W_1 x)$).

1.5.3 The Role of Activation Functions

An **activation function** is a mathematical “gate” applied to the output of a layer ($z = Wx + b$) before passing it to the next layer. Its primary role is to introduce **non-linearity** into the network.

Note

Why do we need Non-Linearity? If we only stacked linear layers without activation functions, the entire deep network would mathematically collapse into a single linear transformation. For example, a 2-layer linear network $f = W_2(W_1 x)$ can be rewritten as $f = W'x$ where $W' = W_2 W_1$. Non-linear activations allow the network to learn and approximate highly complex, non-linear boundaries.

Some of the most common activation functions include:

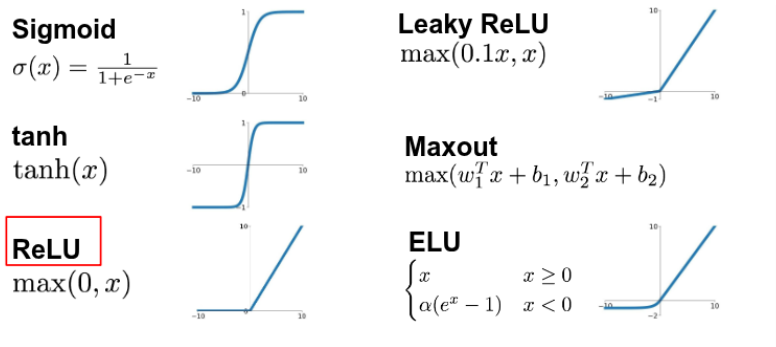


Figure 1.4: Illustration Activation Function.

- **Sigmoid:** $\sigma(x) = \frac{1}{1+e^{-x}}$. It squashes numbers to the range $[0, 1]$. Historically popular, but suffers from the *vanishing gradient* problem.
- **tanh:** Squashes numbers to the range $[-1, 1]$. Like sigmoid, its gradients vanish for very large or very small inputs.
- **ReLU (Rectified Linear Unit):** $f(x) = \max(0, x)$. It is the most widely used default choice because it is computationally efficient and greatly mitigates the vanishing gradient problem.
- **Leaky ReLU and ELU:** Variations of ReLU designed to allow a small, non-zero gradient when $x < 0$ to prevent “dead neurons”.

1.6 Recap on Convolutional Neural Networks (CNNs)

1.6.1 Motivation: The Problem with Fully Connected Layers

In a standard Fully Connected (Dense) layer, every input neuron is connected to every output neuron. If we process a high-resolution image for example, a 1000×1000 RGB image (3×10^6 inputs) even a modest layer with 10^3 hidden units would require over 3 billion parameters.

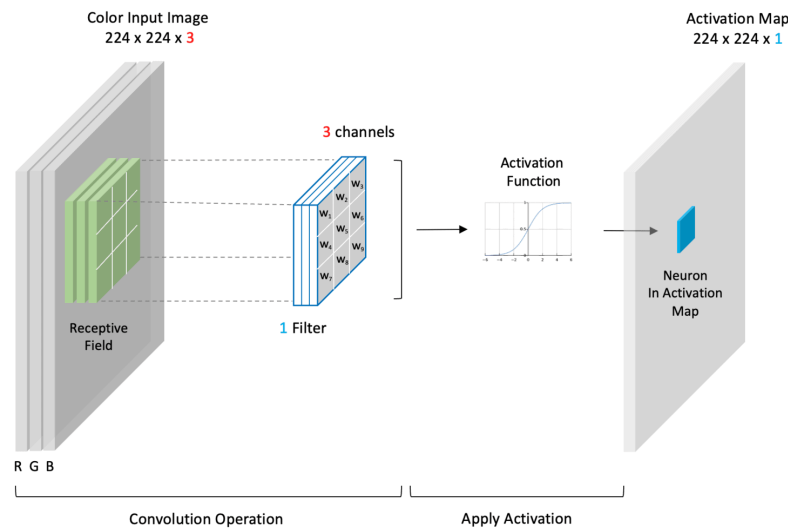


Figure 1.5: Illustration of a single convolutional layer.

This huge parameter count leads to two main issues:

- **Computational Inefficiency & Overfitting:** Storing and updating billions of weights requires massive memory and easily leads to severe overfitting.
- **Loss of Spatial Structure:** Flattening a 2D/3D image into a 1D vector completely ignores the grid-like topology of pixels.

1.6.2 Exploiting Image Structure: Locality and Stationarity

To overcome these limitations, Convolutional Neural Networks (CNN) build upon two fundamental spatial properties of images:

1. **Local Spatial Correlation:** Nearby pixels are strongly related, whereas distant pixels are less relevant. Instead of connecting a unit to the whole image, we can connect it only to a small local patch (a *receptive field*).
2. **Stationarity (Translation Invariance):** Features are location-independent. An edge, texture, or pattern detected in the top-left corner is equally meaningful in the bottom-right corner. Therefore, we can share the same set of weights across all locations.

1.6.3 The Core Building Block: A Simple Convolutional Layer

A Convolutional Layer replaces the dense matrix multiplication with a local, sliding operation.

- **The Kernel (Filter):** A small matrix of learnable weights (e.g., 3×3 or 5×5). It acts as a pattern detector.
- **The Sliding Window:** The kernel slides across the input image step-by-step. At each position, it computes an element-wise multiplication with the underlying image patch and sums the results up.
- **The Feature Map:** The output grid produced by sliding the kernel, Figure 1.6. High values in the feature map indicate a strong match between the kernel's pattern and that specific image region.

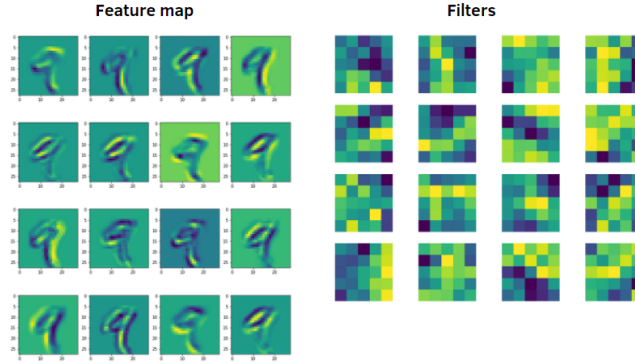


Figure 1.6: Illustration of multiple features map obtained by different filters

1.6.4 Key Hyperparameters: Controlling the Convolution

The behavior and output size of a convolutional layer depend on three primary parameters:

- **Stride (S):** The step size with which the kernel moves across the image. A stride of 1 moves pixel-by-pixel, while $S > 1$ downsamples the output.
- **Padding (P):** Adding zero-valued borders around the input image to preserve spatial dimensions and process edge pixels effectively.
- **Input/Output Channels (C):** Real images have depth (e.g., $C_{in} = 3$ for RGB). A kernel must match the depth of the input, resulting in a 3D tensor of shape $(C_{in} \times K_h \times K_w)$. Using C_{out} distinct kernels produces C_{out} output feature maps.

1.6.5 Formal Mathematical Definition

Mathematically, the operation performed on a 2D input matrix I by a 2D kernel g to yield a feature map f at position $[m, n]$ is defined as:

$$f[m, n] = (I * g)[m, n] = \sum_k \sum_l I[m - k, n - l] g[k, l] \quad (1.1)$$

where k and l iterate over the spatial dimensions of the kernel.

Note on Deep Learning vs. Signal Processing:

In classic signal processing, the index subtraction $I[m - k, n - l]$ implies flipping the filter g by 180° .

However, deep learning frameworks (like PyTorch and TensorFlow) compute **Cross-Correlation** instead:

$$f[m, n] = (I \star g)[m, n] = \sum_k \sum_l I[m + k, n + l] g[k, l] \quad (1.2)$$

Skipping the flipping step saves computation. Since kernel weights $g[k, l]$ are learned from scratch via back-propagation, the network simply learns the pre-flipped weights directly, making the distinction mathematically equivalent for optimization.

1.6.6 Intuitive Examples: Classic Image Processing Filters

Before moving to 3D volumes, let us observe how simple 2D matrices (kernels) process local pixel neighborhoods to extract features or alter images.

Example

1. Smoothing Filter (Noise Reduction):

Replacing each pixel with a weighted combination of its local neighborhood dampens noise and sharp anomalies:

$$\begin{bmatrix} 2 & 3 & 3 \\ 3 & \mathbf{20} & 2 \\ 3 & 2 & 3 \end{bmatrix} \xrightarrow{\text{Smoothing}} \begin{bmatrix} 2 & 3 & 3 \\ 3 & \mathbf{3} & 2 \\ 3 & 2 & 3 \end{bmatrix}$$

The central outlier value (20) is effectively smoothed down to 3.

Example

2. Vertical Edge Detection Filter:

Filters with opposing positive and negative weights capture spatial variations in intensity. For example, a vertical edge detector is defined as:

$$g_{\text{vertical}} = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

Note: Transposing g_{vertical} yields a horizontal edge detector $g_{\text{horizontal}} = g_{\text{vertical}}^T$.

Example

3. Step-by-Step 2D Convolution Example (Stride 1, Pad 1):

Consider convolving a 3×3 image I with a 3×3 vertical edge filter g , applying **Padding** $P = 1$ and **Stride** $S = 1$:

$$\begin{bmatrix} g[k,l] \\ -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} I[k,l] \\ 8 & 5 & 2 \\ 7 & 5 & 3 \\ 9 & 4 & 1 \end{bmatrix} = \begin{bmatrix} f[m,n] \\ -10 & 10 & 10 \\ -14 & 18 & 14 \\ -9 & 12 & 9 \end{bmatrix}$$

Calculation Detail (Accounting for 180° Kernel Flip):

According to $f[m,n] = \sum I[m-k, n-l]g[k,l]$, the kernel g is flipped by 180° prior to element-wise multiplication:

$$g_{\text{flipped}} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

- **Center Value** $f[1, 1] = 18$: Flipped kernel overlaps fully with I :

$$(1 \cdot 8 - 1 \cdot 2) + (1 \cdot 7 - 1 \cdot 3) + (1 \cdot 9 - 1 \cdot 1) = 6 + 4 + 8 = 18$$

- **Top-Left Value** $f[0, 0] = -10$: Flipped kernel overlaps with top-left corner of zero-padded image:

$$(1 \cdot 0 - 1 \cdot 5) + (1 \cdot 0 - 1 \cdot 5) = -10$$

1.6.7 Output Shape and Parameter Counting for 3D Volumes

In practical deep learning architectures, inputs are 3D tensors ($C_{\text{in}} \times H \times W$) and layers use multiple filters to extract distinct visual features.

Spatial Output Size Formula

Given an input of spatial size W , kernel size K , padding P , and stride S , the output dimension W_{out} is given by:

$$W_{\text{out}} = \left\lfloor \frac{W - K + 2P}{S} \right\rfloor + 1 \quad (1.3)$$

Example

Example: Output Volume Size & Parameters Calculation

If the input volume is a $32 \times 32 \times 3$ image, and we apply **10 filters of size 5×5** (with Stride $S = 1$, Padding $P = 2$):

- **Output spatial size:** $(32 + 2 \cdot 2 - 5) / 1 + 1 = 32$ spatially (32×32).
- **Output volume size:** $32 \times 32 \times 10$.
- **Number of parameters:** Each filter has $5 \times 5 \times 3 + 1$ (bias) = 76 parameters. For 10 filters: $76 \times 10 = \mathbf{760}$ parameters.

1.6.8 The Pooling Layer

A Convolutional Neural Network usually interleaves Convolutional layers with Pooling layers.

Purpose of Pooling:

The primary goal of a pooling layer is to perform **spatial downsampling** (subsampling). By doing so, it:

- **Reduces the spatial dimensions** (width and height) of the representation, making it smaller and more computationally manageable.
- **Reduces the number of parameters** in the subsequent fully connected layers, thus helping to control overfitting.
- **Provides Translation Invariance:** Small shifts or distortions in the input image will not drastically change the pooled output, making the network more robust to exact feature locations.

How it Operates:

Unlike the convolution layer, a pooling layer **operates independently on every depth slice** (activation map) of the input. It does not mix channels. If the input has C channels, the output will also have C channels.

Max Pooling

The most common implementation is **Max Pooling**. Instead of computing a dot product like a filter, it simply slides a window across the image and takes the **maximum value** within that window.

Example**Max Pooling Example (Stride 2, Filter 2×2):**

Consider a single depth slice (activation map) of size 4×4 . If we apply a Max Pooling layer with a 2×2 filter and a stride of 2:

1	1	2	4
5	6	7	8
3	2	1	0
1	2	3	4

 $\xrightarrow{\text{Max Pool } 2 \times 2, S=2}$

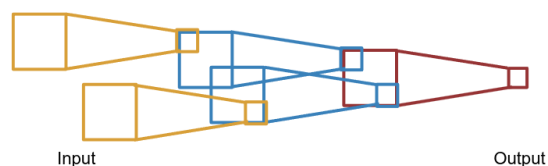
6	8
3	4

The spatial dimensions are exactly halved. The operation discards 75% of the activations, keeping only the most prominent features (highest numbers) in each local 2×2 region.

1.6.9 Visualizing ConvNets**Key Insight**

Understanding what deep CNNs learn requires visualization techniques (e.g., Deconvolutional Networks by Zeiler and Fergus). By projecting high-level feature maps back to the pixel space:

- **Layer 1:** Learns low-level features like edges and colors.
- **Layer 2:** Learns simple textures and geometric patterns.
- **Layer 3 & Higher:** Learns complex, object-specific parts (e.g., dog faces, car wheels) and eventually full objects.



Bibliography

- [1] Fabio Galasso. *Recap on Image Classification and Neural Networks*. Advanced Machine Learning Course Slides, Sapienza Università di Roma. Sept. 2025.